

STA3000F: Advanced Theory of Statistics

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1 Logistics

- Course website: mouwenlong.github.io/teaching/sta3000f26
- Grading: take-home exam + project

1.1 Outline

- Basic probability theory review (concentration inequalities)
- Decision theory (minimax, Bayes, admissibility)
- Modern tools (empirical processes)
- Basics of nonparametric statistics (density estimation, regression)

2 Probability review

2.1 Radon–Nikodym derivative

Idea: generalize probability density function to general measures.

- μ measure of interest
- ν reference measure

We say μ is *absolutely continuous* with respect to ν if $\nu(A) = 0 \implies \mu(A) = 0$. In this case, there exists a function f such that

$$f(x) = \frac{d\mu}{d\nu}(x) \quad \text{for } \nu\text{-almost every } x.$$

For any measurable set A , we have

$$\mu(A) = \int_A f(x) d\nu(x).$$

2.2 Basic probability inequalities

Markov inequality For a non-negative random variable X and any $a > 0$,

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}.$$

Proof:

$$\mathbb{P}(X \geq a) = \mathbb{E}[\mathbf{1}_{X \geq a}] \leq \mathbb{E}\left[\frac{X}{a} \mathbf{1}_{X \geq a}\right] \leq \frac{\mathbb{E}[X]}{a}.$$

Remark: as $a \rightarrow \infty$, the tail probability $\mathbb{P}(X \geq a)$ decays at least as fast as $1/a$. This is a very weak bound.

Chebyshev inequality: For a random variable X with finite mean μ and finite non-zero variance σ^2 , and any $a > 0$,

$$\mathbb{P}(|X - \mu| \geq a) \leq \frac{\sigma^2}{a^2}.$$

Proof: by Markov inequality, we have

$$\mathbb{P}(|X - \mu| \geq a) = \mathbb{E}[\mathbf{1}_{|X - \mu| \geq a}] \leq \mathbb{E}\left[\frac{(X - \mu)^2}{a^2} \mathbf{1}_{|X - \mu| \geq a}\right] \leq \frac{\mathbb{E}[(X - \mu)^2]}{a^2} = \frac{\sigma^2}{a^2}.$$

Remark: second moment gives a better bound than the first moment. As $a \rightarrow \infty$, the tail probability $\mathbb{P}(|X - \mu| \geq a)$ decays at least as fast as $1/a^2$.

Higher moments? Suppose that the random variable X has finite k -th moment $\mathbb{E}[|X|^k] < \infty$ for some $k > 0$. Then, for any $a > 0$,

$$\mathbb{P}(|X - \mathbb{E}[X]| \geq a) \leq \frac{\mathbb{E}[|X - \mathbb{E}[X]|^k]}{a^k}.$$

Can we get a faster decay? Exponential decay?

Moment generating function (MGF): For a random variable X , the moment generating function is defined as

$$m_X(t) = \mathbb{E}[e^{tX}], \quad t \in \mathbb{R}.$$

Why is it called “moment generating function”? Taylor expansion of $m_X(t)$ at $t = 0$ gives

$$m_X(t) = \mathbb{E}[e^{tX}] = \sum_{k=0}^{\infty} \frac{t^k}{k!} \mathbb{E}[X^k].$$

So if you take higher order derivatives of $m_X(t)$ at $t = 0$, you can get the moments of X .

- $X \sim \mathcal{N}(\mu, \sigma^2)$, then $m_X(t) = e^{\mu t + \frac{1}{2}\sigma^2 t^2}$.
- $X \sim \text{Bernoulli}(p)$, then $m_X(t) = 1 - p + pe^t$.

Note: moment generating function may not exist for all $t \in \mathbb{R}$, e.g., Cauchy distribution (moment does not even exist); exponential distribution (moment generating function only exists for $t < \lambda$).

By applying Markov inequality to $e^{t(X - \mathbb{E}[X])}$, we have

$$\mathbb{P}(X - \mathbb{E}[X] \geq a) = \mathbb{P}(e^{t(X - \mathbb{E}[X])} \geq e^{ta}) \leq \frac{\mathbb{E}[e^{t(X - \mathbb{E}[X])}]}{e^{ta}} = e^{-ta} m_X(t) e^{-t\mathbb{E}[X]}.$$

Note: this holds for any $t > 0$. So we can take infimum over $t > 0$ to get the best bound.

Example 1. For $X \sim \mathcal{N}(\mu, \sigma^2)$, we have

$$\mathbb{P}(X \geq \mu + a) \leq e^{-ta} m_X(t) e^{-t\mu} = e^{-ta} e^{\mu t + \frac{1}{2}\sigma^2 t^2} e^{-t\mu} = e^{-ta + \frac{1}{2}\sigma^2 t^2}.$$

Take the optimal choice of $t = a/\sigma^2$, we have

$$\mathbb{P}(X \geq \mu + a) \leq e^{-\frac{a^2}{2\sigma^2}}.$$

For normal distribution, we know the exact tail probability. But for more general distributions, we can use moment generating function to get a bound on the tail probability.

Definition 1. X is called sub-Gaussian if there exists a constant $\sigma > 0$ such that for all $t \in \mathbb{R}$,

$$\mathbb{E}[e^{t(X-\mathbb{E}[X])}] \leq e^{\frac{1}{2}\sigma^2 t^2}.$$

e.g. Rademacher random variable X with $\mathbb{P}(X = 1) = \mathbb{P}(X = -1) = 1/2$.

$$\begin{aligned} m_X(t) &= \mathbb{E}[e^{tX}] = \frac{1}{2}e^t + \frac{1}{2}e^{-t} = \cosh(t) \\ &\leq 1 + \frac{t^2}{2} + \frac{t^4}{4!} + \frac{t^6}{6!} + \dots \end{aligned}$$

On the other hand, we have

$$e^{\frac{1}{2}t^2} = 1 + \frac{t^2}{2} + \frac{t^4}{2^2 2!} + \frac{t^6}{2^3 3!} + \dots$$

Term-wise comparison shows that $\cosh(t) \leq e^{\frac{1}{2}t^2}$ for all $t \in \mathbb{R}$. So Rademacher random variable is sub-Gaussian with $\sigma = 1$.

Sum (average) of i.i.d. random variables: Let $X_1, \dots, X_n$ be i.i.d. sub-Gaussian random variables with parameter σ . Can we say something about $\frac{1}{n} \sum_{i=1}^n X_i$?

For any $t \in \mathbb{R}$, we have

$$\mathbb{E}\left[e^{t\left(\frac{1}{n} \sum_{i=1}^n X_i - \mathbb{E}[X_i]\right)}\right] = \left\{\mathbb{E}\left[e^{\frac{t}{n}(X_1 - \mathbb{E}[X_1])}\right]\right\}^n = \left\{m_{X_1}(t/n)\right\}^n \leq \exp\left(\frac{\sigma^2 t^2}{2n}\right).$$

So, the average is $\frac{\sigma}{\sqrt{n}}$ -sub-Gaussian, and consequently,

$$\mathbb{P}\left(\frac{1}{n} \sum_{i=1}^n X_i - \mathbb{E}[X_i] \geq a\right) \leq e^{-ta} m_{X_1 - \mathbb{E}[X_1]}(t/n)^n \leq e^{-ta + \frac{\sigma^2 t^2}{2n}}.$$

Taking the optimal choice of $t = an/\sigma^2$, we have

$$\mathbb{P}\left(\frac{1}{n} \sum_{i=1}^n X_i - \mathbb{E}[X_i] \geq a\right) \leq e^{-\frac{na^2}{2\sigma^2}}.$$

This is called Hoeffding's inequality. It shows that the average of i.i.d. sub-Gaussian random variables concentrates around its mean with exponential decay.

e.g. $X_1, \dots, X_n$ are i.i.d. Rademacher random variables. Then, for any $a > 0$,

$$\mathbb{P}\left(\left|\frac{1}{n} \sum_{i=1}^n X_i\right| \geq a\right) \leq 2e^{-\frac{na^2}{2}},$$

where the factor of 2 comes from the union bound.

In general, we can also use the same idea to study i.i.d. average of bounded random variables. Let $X_1, \dots, X_n$ be i.i.d. random variables with $\mathbb{E}[X_i] = \mu$ and $X_i \in [a, b]$ almost surely. We can show that $X_i - \mu$ is sub-Gaussian with parameter $\sigma = \frac{b-a}{2}$. Then, we have

$$\mathbb{P}\left(\left|\frac{1}{n} \sum_{i=1}^n X_i - \mu\right| \geq u\right) \leq 2 \exp\left(-\frac{2nu^2}{(b-a)^2}\right).$$

Remarks:

- Deviation is of order $O(1/\sqrt{n})$. Tail decay is exponential.
- Independence is needed in general (can be extended when specific dependence structure is known. e.g. martingale difference sequence). Identical distribution can be relaxed in most cases, as long as we have the MGF bound for each random variable.

Union bound

$$\mathbb{P}\left(\bigcup_{i \in \mathcal{I}} A_i\right) \leq \sum_{i \in \mathcal{I}} \mathbb{P}(A_i).$$

Remark:

- Usually, each A_i corresponds to “bad event at certain direction”. If none of them is happening, then we are in a “good event”.
- In many cases, we have exponentially many bad events. That’s why we need the tail probability to decay exponentially fast, so that the union bound still gives a meaningful bound.

Example: probabilistic method in combinatorics. Consider Hamming cube $\{-1, +1\}^n$. Hamming distance between two points $x, y \in \{-1, +1\}^n$ is defined as

$$d_H(x, y) = \sum_{i=1}^n \mathbf{1}_{x_i \neq y_i}.$$

Question: can we pack a lot of points in the Hamming cube such that any two points are far away from each other? Concretely, we want to find a subset $S \subset \{-1, +1\}^n$ such that for any $x, y \in S$, $d_H(x, y) \geq n/4$.

Answer: S can be exponentially large in n .

Let $X_1, X_2, \dots, X_M \stackrel{\text{i.i.d.}}{\sim} \text{Unif}(\{-1, 1\}^n)$. For each fixed i, j pair, we have

$$\mathbb{P}(d_H(X_i, X_j) < n/4) = \mathbb{P}\left(\sum_{k=1}^n \mathbf{1}_{X_{ik} \neq X_{jk}} < n/4\right) \leq \exp\left(-2n(1/4)^2\right) = e^{-n/8},$$

where we used Hoeffding’s inequality for sum of i.i.d. Bernoulli random variables.

Applying union bound, we have

$$\mathbb{P}\left(\exists i < j \text{ such that } d_H(X_i, X_j) < n/4\right) \leq \binom{M}{2} e^{-n/8} \leq \frac{M^2}{2} e^{-n/8}.$$

We just need a “good” configuration to exist. So we want the probability of the bad event to be strictly less than 1. By choosing $M = \lfloor e^{n/16} \rfloor$, we have

$$\mathbb{P}\left(\exists i < j \text{ such that } d_H(X_i, X_j) < n/4\right) \leq \frac{1}{2}.$$

3 Back to statistics – decision theory

Key question: how to compare different statistical procedures? And how to rigorously define “optimal” procedure? This requires a unified framework.

To start with, we define

- A statistical model $\mathcal{P} = \{P_\theta : \theta \in \Theta\}$, where Θ is the parameter space (here the “parameter” does not have to be finite-dimensional).
- Observe $X \sim P_{\theta^*}$ for some unknown $\theta^* \in \Theta$. Goal: to make decision about θ^* based on X .
- Decision rule $\delta : \mathcal{X} \rightarrow \mathcal{A}$, where $\mathcal{A}$ is the action space. For example, if we want to estimate θ^* , then $\mathcal{A} = \Theta$; for testing hypothesis, $\mathcal{A} = \{0, 1\}$.
- Loss function $L : \Theta \times \mathcal{A} \rightarrow \mathbb{R}_+$, e.g., $L(\theta, a) = \|\theta - a\|^2$ for squared error loss; 0-1 loss for testing hypothesis.
- Risk function: expected loss of a decision rule δ when data is generated from P_θ :

$$R(\theta; \delta) := \mathbb{E}_{X \sim P_\theta}[L(\theta, \delta(X))].$$

e.g. for estimation, consider a known functional $g : \Theta \rightarrow \mathbb{R}^d$. A decision rule δ is an estimator of $g(\theta^*)$. The risk function is

$$R(\theta; \delta) = \mathbb{E}_{X \sim P_\theta}[\|\delta(X) - g(\theta)\|^2].$$

e.g. for testing, we partition the parameter space into two disjoint sets Θ_0 and Θ_1 . The null hypothesis is $H_0 : \theta^* \in \Theta_0$ and the alternative hypothesis is $H_1 : \theta^* \in \Theta_1$. A decision rule δ is a test of the hypothesis, where $\delta(X) = 0$ means we accept H_0 and $\delta(X) = 1$ means we reject H_0 . The risk function is

$$R(\theta; \delta) = \mathbb{E}_{X \sim P_\theta}[\mathbf{1}_{\{\delta(X) \neq \mathbf{1}_{\{\theta \in \Theta_1\}}\}}] = \begin{cases} \mathbb{P}_{X \sim P_\theta}(\text{reject } H_0), & \text{if } \theta \in \Theta_0, \\ \mathbb{P}_{X \sim P_\theta}(\text{accept } H_0), & \text{if } \theta \in \Theta_1. \end{cases}$$

- Smaller risk is better.
- Risk is a function of θ . This becomes a multi-objective optimization problem.
- We may not be able to find a decision rule that is uniformly better than all other decision rules. So we need to define some optimality criteria.

3.1 Admissibility

Definition 2. Let δ be a decision rule. We say δ is inadmissible if there exists another decision rule $\tilde{\delta}$ such that

$$R(\theta; \tilde{\delta}) \leq R(\theta; \delta) \quad \text{for all } \theta \in \Theta,$$

and the inequality is strict for at least one $\theta \in \Theta$. Otherwise, we say δ is admissible.

Problem 1: stupid estimators can also be admissible.

e.g. $X_1, X_2, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\theta, 1)$, $\theta \in \mathbb{R}$. Consider $\hat{\theta}(X) = 0$. Always output 0 regardless of the data. Then, the risk function is

$$R(\theta; \hat{\theta}) = \mathbb{E}_{X \sim P_\theta}[(\hat{\theta}(X) - \theta)^2] = \theta^2.$$

The risk of this estimator at 0 is 0. So any other estimator $\hat{\theta}'$ that dominates $\hat{\theta}$ must have $R(0; \hat{\theta}') = 0$. This means that $\hat{\theta}'(X) = 0$ almost surely when $\theta = 0$. So $\hat{\theta}'$ cannot dominate $\hat{\theta}$. Therefore, $\hat{\theta}$ is admissible.

Problem 2: “good” estimators can be inadmissible.

“Stein’s paradox”: $X_1, X_2, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\theta, I_d)$, $\theta \in \mathbb{R}^d$. Consider a simple estimator

$$\hat{\theta}_n := \frac{1}{n} \sum_{i=1}^n X_i.$$

Theorem 1 (Stein’s paradox). *If $d \geq 3$, then $\hat{\theta}_n$ is inadmissible under squared error loss.*

Proof: by construction. James–Stein estimator:

$$\hat{\theta}_{\text{JS}} = \left(1 - \frac{(d-2)}{n \|\bar{X}_n\|_2^2}\right) \bar{X}_n.$$

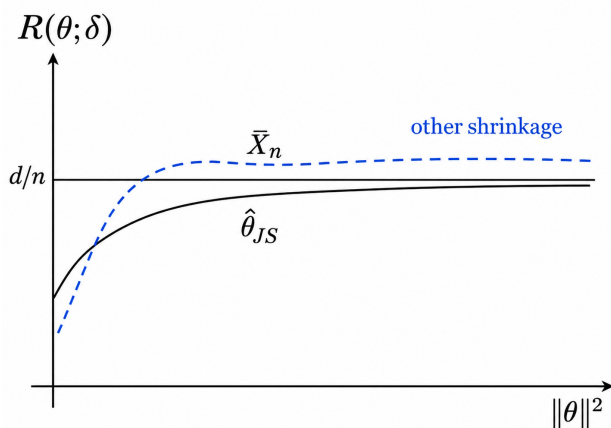
We can calculate

$$\mathbb{E}_\theta [\|\hat{\theta}_{\text{JS}} - \theta\|_2^2] = \frac{d}{n} - \mathbb{E} \left[\left(\frac{(d-2)}{n \|\bar{X}_n\|_2^2} \right)^2 \right] < \frac{d}{n} = \mathbb{E}_\theta [\|\bar{X}_n - \theta\|_2^2].$$

This proves the inadmissibility of $\bar{X}_n$.

Remarks:

- Sample mean is still “good” in many senses, e.g., it is minimax optimal. Even being inadmissible, the gap between sample mean and JS estimator is diminishing fast when $\|\theta\|_2$ becomes large. In practice, it’s not always making a big difference.
- JS estimator itself is also inadmissible, e.g. clip the factor when it goes below 0.
- JS estimator itself is useful. It shrinks towards 0 adaptively based on the data you see. So shrinkage improves the risk at 0 without hurting the risk at non-zero. Related to modern topics including empirical Bayes estimation.



3.2 Bayes criterion

Warning: most of this class is based on a frequentist perspective. But we will also discuss some Bayesian ideas. Here, the prior should be interpreted more as a weighting function on the parameter space, rather than a subjective belief about the parameter.

Definition 3. Given a prior distribution π on Θ , the Bayes risk of a decision rule δ is defined as

$$r_\pi(\delta) := \int_{\Theta} R(\theta; \delta) d\pi(\theta) = \mathbb{E}_{\theta \sim \pi}[R(\theta; \delta)].$$

This becomes a single-objective optimization problem. A decision rule δ_π is called a Bayes rule with respect to prior π if it minimizes the Bayes risk:

$$r_\pi(\delta_\pi) = \inf_{\delta} r_\pi(\delta).$$

How to find the Bayes rule? Suppose that $p_\theta(x) = \frac{dP_\theta}{d\lambda}(x)$ is the density of P_θ with respect to a reference measure λ . Then, we have

$$r_\pi(\delta) = \int_{\Theta} \int_{\mathcal{X}} L(\theta, \delta(x)) p_\theta(x) d\lambda(x) d\pi(\theta)$$

(Fubini–Tonelli) $\quad \quad \quad \int_{\mathcal{X}} \underbrace{\int_{\Theta} L(\theta, \delta(x)) p_\theta(x) d\pi(\theta)}_{\text{We can pointwise minimize this for each } x} d\lambda(x).$

For each $x \in \mathcal{X}$, we can find $\delta_\pi(x)$ that minimizes the inner integral, i.e.,

$$\delta_\pi(x) = \arg \min_{a \in \mathcal{A}} \int_{\Theta} L(\theta, a) p_\theta(x) d\pi(\theta). \quad (1)$$

In Bayesian statistics, we see $\theta \sim \pi$ as a random variable. Then, the posterior distribution of θ given $X = x$ is

$$d\Pi(\theta | X) = \frac{p_\theta(x) d\pi(\theta)}{\int_{\Theta} p_{\theta'}(x) d\pi(\theta')}.$$

Consider the posterior expected loss:

$$\mathbb{E}_{\theta \sim \Pi(\cdot | X)}[L(\theta, a)] = \int_{\Theta} L(\theta, a) d\Pi(\theta | X) = \frac{\int_{\Theta} L(\theta, a) p_\theta(x) d\pi(\theta)}{\int_{\Theta} p_{\theta'}(x) d\pi(\theta')}.$$

Operationally, this is the conditional expected loss by choosing a after observing $X = x$.

The denominator does not depend on a . So minimizing the posterior expected loss is equivalent to minimizing the numerator, which is exactly the same as (1). So, Bayes rule is equivalent to minimizing the posterior expected loss for each X .

$$\delta_\pi(X) = \arg \min_{a \in \mathcal{A}} \mathbb{E}_{\theta \sim \Pi(\cdot | X)}[L(\theta, a)].$$

For estimation problem with squared loss, i.e., $L(\theta, a) = \|a - g(\theta)\|_2^2$, we have

$$\delta_\pi(X) = \arg \min_{a \in \mathbb{R}^k} \int_{\Theta} \|a - g(\theta)\|_2^2 d\Pi(\theta | X) = \int_{\Theta} g(\theta) d\Pi(\theta | X).$$

So the Bayes estimator is the posterior mean of $g(\theta)$. In practice, we usually use MCMC or variational inference to approximate the posterior distribution and compute the posterior mean.